Use Case	System Operations
Book Appointment	searchPatient(insuranceNumber), requestAppointment(patient, healthcareProfessional, timeSlot, medicalService)
Issue Prescription	selectConsultation(consultation), startPrescription(consultation) searchMedication(name), selectMedication(medication, dose), issuePrescription(instructions)
Register Patient Arrival	searchPatient(insuranceNumber), findAppointment(patient), registerPatientArrival(appointment, arrivalTime)

Contract CO1: searchPatient

Operation: searchPatient(insuranceNumber)

Cross References: Book Appointment, Register Patient Arrival, Issue Prescription

Preconditions: The user is logged in as Authorized Clinic Staff.

Postconditions: None

Contract CO2: requestAppointment

Operation: requestAppointment(patient, healthcareProfessional, timeSlot, medicalService)

Cross References: Book Appointment

Preconditions:

- User is logged in as authorized clinic staff
- patient exists in PatientCatalog
- timeSlot belongs to healthcareProfessional's Schedule with status available
- If medicalService requires a referral, a valid Referral exists

Postconditions:

- An `Appointment` instance `a` was created. (instance creation)
- `a.status` became "Requested". (attribute modification)
- `a.healthcareProfessional` became `healthcareProfessional`. (attribute modification)
- `timeSlot.status` became "Booked". (attribute modification)
- `a` was associated with `patient`. (association formed)
- `a` was associated with `timeSlot`. (association formed)
- `a` was associated with `medicalService`. (association formed)
- `a` was associated with the requesting Authorized Clinic Staff. (association formed)
- If a referral was required, `a` was associated with the `Referral`. (association formed)
- `a` was added to the `AppointmentCatalog`. (association formed)

Contract CO3: selectConsultation

Operation: selectConsultation(consultation)

Cross References: Start Prescription, Issue Prescription, Record Clinical Information

Preconditions:

- The user is logged in as Healthcare Professional
- consultation exists

Postconditions: None

Contract CO4: startPrescription

Operation: startPrescription(consultation)

Cross References: Issue Prescription

Preconditions:

- The user is logged in as Healthcare Professional
- consultation exists with status active

Postconditions:

- A Prescription instance p was created (instance creation)
- p.id was set to a unique integer (attribute modification)
- p.status was set to "Draft" (attribute modification)
- p was associated with consultation (association formed)
- p was associated with PrescriptionCatalog (association formed)

Contract CO4: searchMedication

Operation: searchMedication(name)

Cross References: Issue Prescription

Preconditions: The user is logged in as Healthcare Professional

Postconditions: None

Contract CO5: selectMedication

Operation: selectMedication(medication, dose)

Cross References: Issue Prescription

Preconditions:

- The user is logged in as Healthcare Professional
- A Prescription p with status "Draft" exists
- medication exists in MedicationCatalog

Postconditions:

- A PrescriptionLine instance pl was created (instance creation)
- pl was associated with medication (association formed)
- pl.dose became dose (attribute modification)
- pl was associated with p (association formed)

Contract CO6: issuePrescription

Operation: issuePrescription(instructions)

Cross References: Start Prescription

Preconditions:

- The user is logged in as Healthcare Professional
- A Prescription p with status "Draft" exists

Postconditions:

- p.instructions became instructions (attribute modification)
- p.status became "Issued" (attribute modification)
  p.dateIssued became the current date (attribute modification)

Contract CO7: findAppointment
Operation: findAppointment(patient)

Cross References: Register Patient Arrival
Preconditions:

- The user is logged in as Administrative Staff
- patient exists in PatientCatalog

Postconditions: None

Contract CO8: registerPatientArrival
Operation: registerPatientArrival(appointment, arrivalTime)

Cross References: Register Patient Arrival
Preconditions:

- The user is logged in as Administrative Staff
- appointment exists in AppointmentCatalog

Postconditions:

- appointment.arrivalTime was changed to arrivalTime (attribute modification)